SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO1: Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 20%

QUESTIONS

Q.No	Question	Bloom's Level
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

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Code of Conduct:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

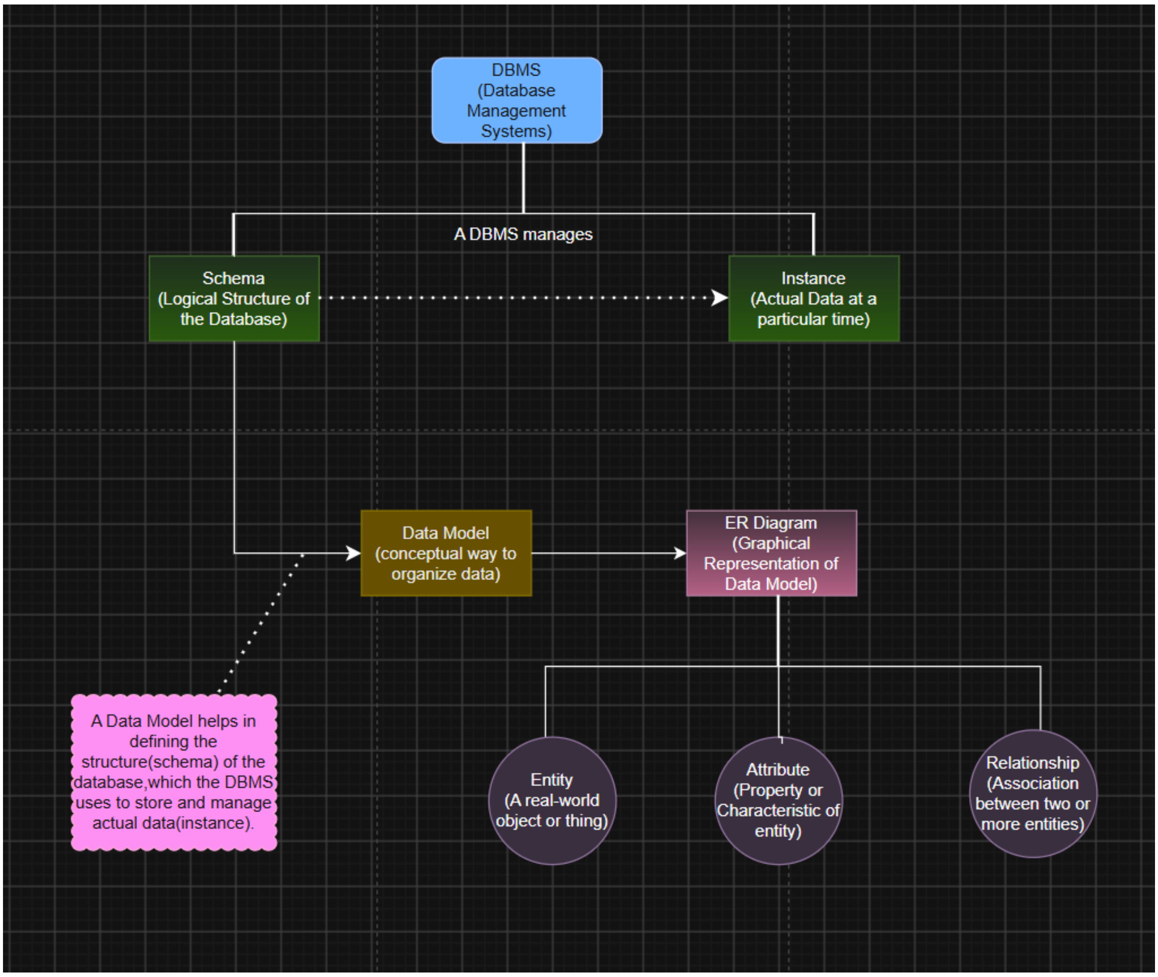
I **R.Pranathi(192525076**) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Pranathi**

RUBRICS FOR CALCULATION:

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. **Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.**



DBMS (Database Management System)

- Software used to create, store, retrieve, and manage databases.
- Ensures data security, consistency, and integrity.

Schema

- The logical structure or blueprint of a database.
- Defines tables, columns, constraints, and relationships.
- Changes rarely.

Instance

- The actual data stored in the database at a particular time.
- Changes whenever data is inserted, updated, or deleted.

Data Model

- A set of concepts used to organize and describe data.
- Defines how data is stored and related.
- Examples: Relational Model, Hierarchical Model.

ER Diagram (Entity-Relationship Diagram)

- A graphical representation of a database.
- Shows entities, attributes, and relationships.

- Used during database design.

Entity

- A real-world object or thing.
- Examples: Student, Employee, Customer.

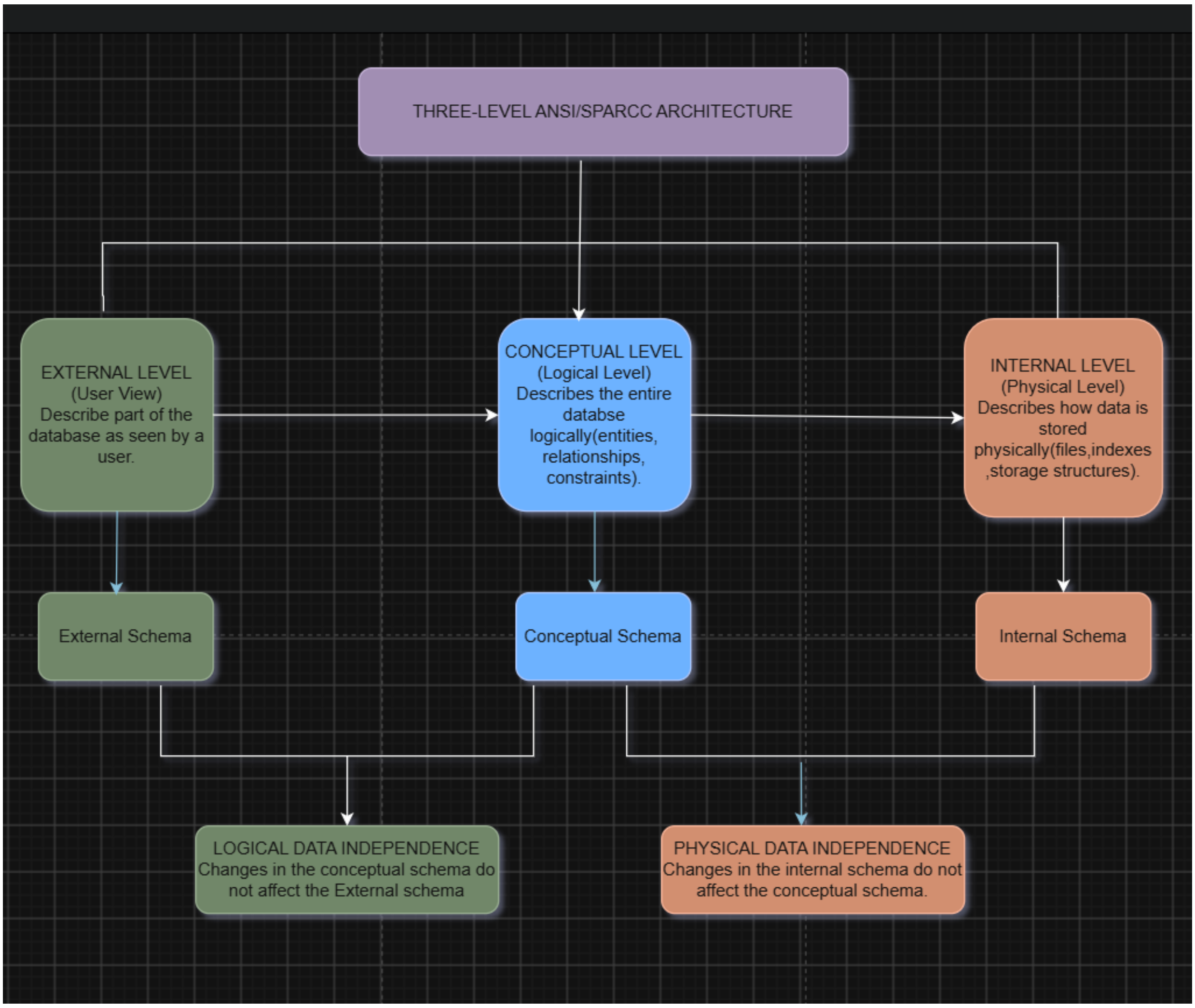
Attribute

- A property or characteristic of an entity.
- Examples: Student_ID, Name, Age.

Relationship

- Represents the association between two or more entities.
- Example: **Student** enrolls in **Course**.

2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.



External Level (User View)

- Represents different views for different users.
- Shows only the data required by each user.
- Provides security by hiding unnecessary information.

Conceptual Level (Logical View)

- Describes the complete logical structure of the database.
- Defines entities, relationships, attributes, and constraints.
- Independent of physical storage details.

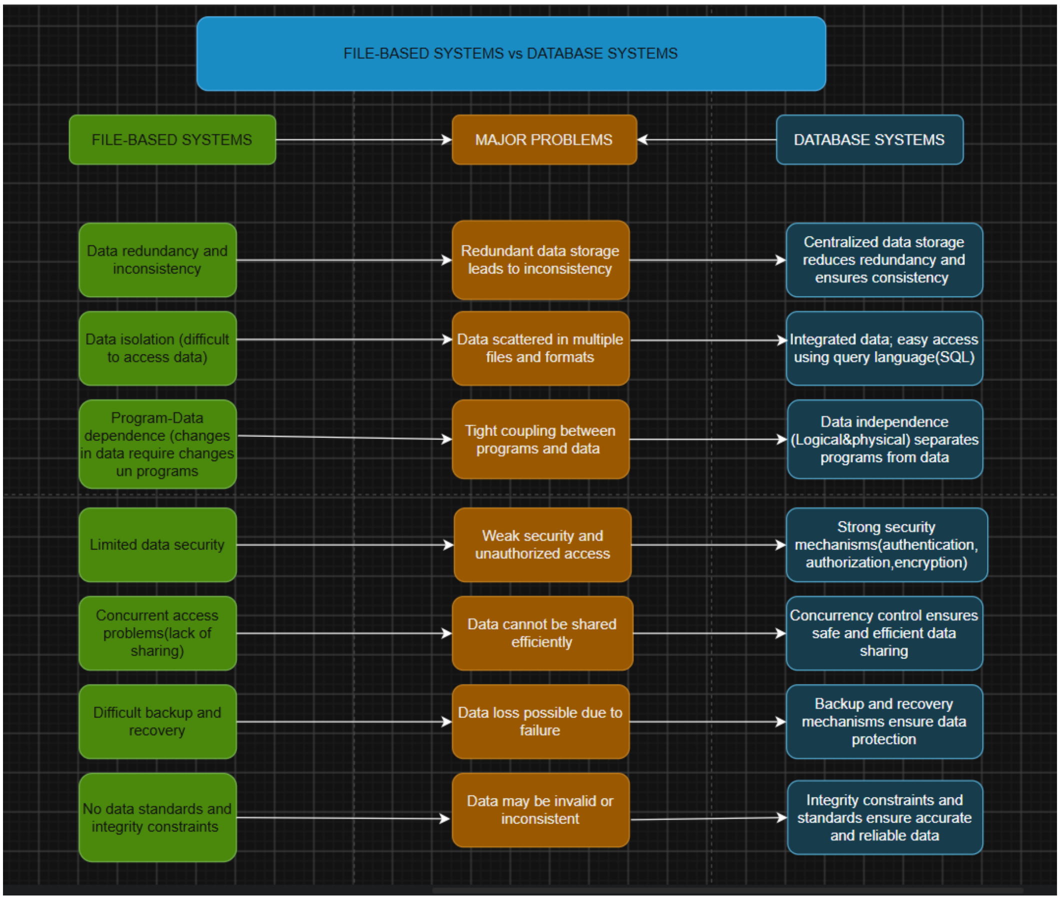
Internal Level (Physical View)

- Describes how data is physically stored.
- Includes file organization, indexes, and storage methods.
- Focuses on database performance and storage efficiency.

Data Independence

- **Logical Data Independence:** Changes in the conceptual schema do not affect external user views.
- **Physical Data Independence:** Changes in the internal schema do not affect the conceptual schema.

3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS



File-Based Systems

- Stores data in separate files.
- Leads to data redundancy and inconsistency.
- Difficult to access, update, and share data.
- Limited security and backup facilities.
- No data independence.

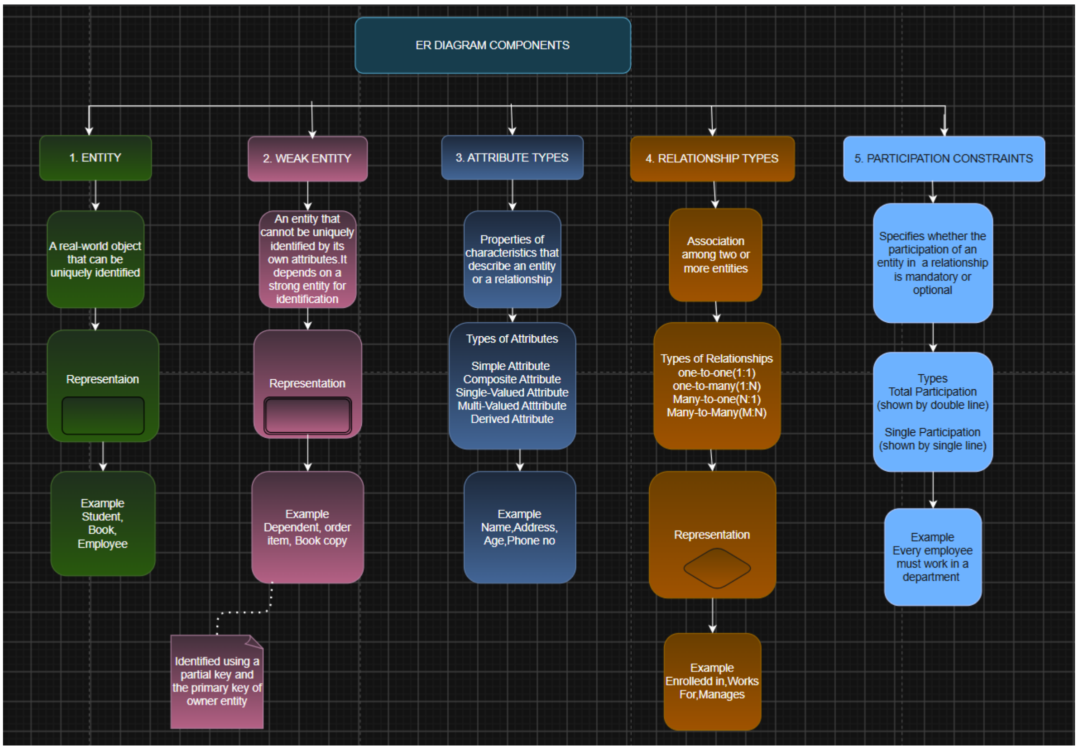
Database Systems (DBMS)

- Stores data in a centralized database.
- Reduces data redundancy and improves consistency.
- Provides easy data access using SQL.
- Supports security, backup, and recovery.
- Offers logical and physical data independence.

Problems Solved by DBMS

- Eliminates data redundancy and inconsistency.
- Integrates data into a single database.
- Provides efficient data retrieval.
- Controls concurrent access by multiple users.
- Enhances data security.
- Supports backup and recover

4. **Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.**



Entity

- A real-world object or thing that can be identified.
- Represented by a Rectangle.
- Examples: Student, Employee, Department.

Weak Entity

- An entity that cannot be uniquely identified by its own attributes.
- Depends on a strong (owner) entity.
- Represented by a Double Rectangle.
- Example: Dependent (depends on Employee).

Attribute Types

- Simple Attribute: Cannot be divided (e.g., Age).
- Composite Attribute: Can be divided into sub-parts (e.g., Address → Street, City, PIN).
- Single-Valued Attribute: Has only one value (e.g., Roll Number).
- Multi-Valued Attribute: Has multiple values (e.g., Phone Number).
- Derived Attribute: Obtained from another attribute (e.g., Age from Date of Birth).
- Key Attribute: Uniquely identifies an entity (e.g., Student_ID)

Relationship Types

- Shows the association between two or more entities.
- Represented by a Diamond.
- Types:
    - One-to-One (1:1)
    - One-to-Many (1:N)
    - Many-to-Many (M:N)

Participation Constraints

- Specify whether an entity must participate in a relationship.
- Total Participation: Every entity must participate (Mandatory).
- Partial Participation: Participation is optional.5. Construct a concept map comparing Hierarchical, Network, and Relationaal data models with their key properties.

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Hierarchical Data Model

- Organizes data in a **tree structure**.
- Each child has only one parent.
- Supports **one-to-many (1:M)** relationships.
- Simple but less flexible.

Network Data Model

- Organizes data in a **graph structure**.
- A child can have multiple parents.
- Supports **many-to-many (M:N)** relationships.
- More flexible than the hierarchical model.

Relational Data Model

- Organizes data in **tables (rows and columns)**.
- Tables are connected using **primary and foreign keys**.
- Supports SQL for easy data access.
- Most widely used data model in modern times.